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DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION ENGINEERING

ACADEMIC YEAR: 2025-26 (Even sem)

PROJECT PROPOSAL

Project Title: FPGA Based Automatic Irrigation System

Department: Electronics and Telecommunication Engineering

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Project Guide Remarks:

Prof. Gauri Vaidya
Project Guide



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Project Title: FPGA Based Automatic Irrigation System

Abstract: This project presents the design and simulation of an FPGA-Based Automatic Irrigation System using the JXCT 7-in-1 Soil Sensor (Modbus RS485, 5V). The Xilinx XC7A35T Artix-7 FPGA acts as the central controller, processing seven soil parameters — moisture, temperature, electrical conductivity (EC), pH, Nitrogen (N), Phosphorus (P), and Potassium (K) — received digitally via Modbus RTU RS485 protocol. A Finite State Machine (FSM) implemented in Verilog HDL evaluates all parameters simultaneously and activates the water pump only when soil is dry and all safety conditions are met. A custom web-based FPGA simulator and EDA Playground waveform simulation have been used for complete design validation.

Background and Rationale: Agriculture consumes approximately 80% of India's freshwater, yet 40–60% is wasted due to inefficient manual irrigation. Conventional single-parameter moisture sensors cannot account for soil health factors like salinity (EC), pH, and nutrient levels. An FPGA-based system using an industrial 7-in-1 sensor addresses this by enabling real-time multi-parameter soil monitoring with hardware-level speed, deterministic response, and no software dependency — making it significantly superior to microcontroller-based alternatives for precision agriculture.

Objectives:

1. Monitor seven soil parameters (moisture, temperature, EC, pH, N, P, K) simultaneously in real time via Modbus RS485.



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2. Implement FSM-based combinational logic in Verilog HDL on Xilinx XC7A35T FPGA for intelligent pump control.
3. Activate water pump only when soil is dry AND temperature, EC, and pH are within safe thresholds.
4. Generate multi-parameter nutrient deficiency alerts using N, P, K sensor readings.
5. Validate complete design through EDA Playground HDL simulation, Tinkercad circuit simulation, and custom FPGA web simulator.

METHODOLOGY: The JXCT 7-in-1 sensor transmits all seven parameters in a single Modbus RTU RS485 frame. The FPGA's UART module decodes the 25-byte response. Nine pre-set threshold constants (localparam) embedded in FPGA logic fabric are compared against incoming values using combinational assign statements. A 4-state FSM (IDLE → READ_SENSOR → PROCESS → OUTPUT) cycles continuously. In the OUTPUT state, the system checks: (1) safe_to_irrigate = temp_safe AND ec_safe AND ph_safe; (2) if soil_dry AND safe_to_irrigate → pump ON, green LED ON; (3) else if soil_wet → pump OFF, red LED ON; (4) if nutrients_ok = FALSE → alert ON. The pump is driven via an NPN BC547 transistor with 220Ω base resistor. Simulation was performed on EDA Playground (Icarus Verilog + EPWave) and a custom browser-based FPGA simulator with live waveform output.



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Expected Outcomes: A fully synthesizable Verilog HDL design verified by simulation, demonstrating automatic pump control based on 7-parameter soil analysis. System achieves 30–50% water saving compared to conventional systems, prevents crop damage from saline/acidic soil irrigation, and provides NPK deficiency alerts. A custom FPGA web simulator provides a printable, interactive demonstration of the system for academic evaluation.

Scalability of the Project: The design is modular and scalable: (1) Multi-zone irrigation using additional FPGA GPIO banks. (2) IoT integration via ESP8266/ESP32 Wi-Fi module for cloud dashboard monitoring. (3) Solar power system for off-grid field deployment. (4) Mobile app control via MQTT protocol. (5) Automated NPK-based fertilizer injection pump. (6) AI-based predictive irrigation using FPGA DSP slices for on-device neural inference.

Market Analysis: The global smart irrigation market is valued at USD 1.8 billion (2023) and projected to reach USD 5.0 billion by 2030 (CAGR 15.8%). India's precision agriculture market is growing at 12% annually driven by PM-KUSUM, NMSA, and Digital Agriculture Mission schemes. Industrial Modbus RS485 soil sensors like JXCT are used in commercial greenhouse management, smart farms, and government agri-IoT projects — sectors directly addressable by this technology.



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Comparative analysis:

Parameter	This Project (FPGA+7-in-1)	Arduino + Simple Sensor	MCU + Multi Sensor	Manual Irrigation
Parameters Monitored	7 (M, T, EC, pH, N, P, K)	1 (Moisture only)	3–4	0 (Visual only)
Processing Speed	Hardware (clock-level)	Software (ms delay)	Software (ms delay)	N/A
Communication Protocol	Modbus RTU RS485 (Digital)	Analog (noise- prone)	I2C / SPI	None
Safety Logic	EC, pH, Temp safety	None	Partial	None
Nutrient Alert (N,P,K)	Yes	No	No	No
Water Saving	30–50%	10–20%	15–25%	0%

TIMELINE:

Phase	Activity	Deliverable	Timeline
1	System Architecture Design & Verilog HDL Coding	Complete module declaration, FSM, threshold constants, Verilog source file	Jan – Feb 2026
2	HDL Simulation (EDA Playground + EPWave)	Testbench, waveform screenshots, 6 test case verification	Feb – Mar 2026



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3	Web Simulator Development & Tinkercad Simulation	Custom FPGA web simulator, Tinkercad circuit demo, mobile- compatible	Mar 2026
4	Hardware Procurement & Assembly (Post- funding)	JXCT sensor received, MAX485 interface, FPGA bitstream upload	Apr – May 2026
5	Field Testing, PCB Design & Report Finalization	Soil field test results, PCB layout, final project report & viva	May 2026

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Coordinator / Principal Remarks: